# Assignment - 2

1. Check whether a list is empty or not.

Method 1: using if

```
lst = []
if not lst:
    print ("List is empty")
else:
    print ("List is not empty")
```

Method 2: using len()

```
lst = []
if len(lst) == 0:
    print ("Empty")
else:
    print ("not empty")
```

2. Get unique values from a list

Method 1: using set()

```
lst = [1,2,2,3,4,4,5]
unique_vals = list(set(lst))
print (unique_vals)
```

Method 2: without using set()

```
lst = [1,2,2,3,4,4,5]
unique_vals = []
for i in lst:
    if i not in unique_vals:
        unique_vals.append(i)
print (unique_vals)
```

3. Multiply all numbers in a List (3 ways)

way 1 : using for loop :

```
lst = [1, 2, 3, 4]

prod = 1
for i in lst:
    prod *= i
print(prod)
```

way 2 : using math.prod()

```
import math

lst = [1, 2, 3, 4]
print(math.prod(lst))
```

way 3 : using reduce()

```
from functools import reduce

lst = [1, 2, 3, 4]
prod = reduce(lambda x, y : x * y, lst)

print(prod)
```

4. Transpose a matrix in a single line

using zip() :

```
matrix = [[1, 2, 3], [4, 5, 6]]
transpose = list(map(list, zip(*matrix)))
print(transpose)
```

5. multiply two matrices in a single line using numpy.

```
import numpy as np

A = np.array([[1,2],[3,4]])

B = np.array([[5,6],[7,8]])

C = A.dot(B)
    print(C)

(OR)
print(np.matmul(A,B))
```